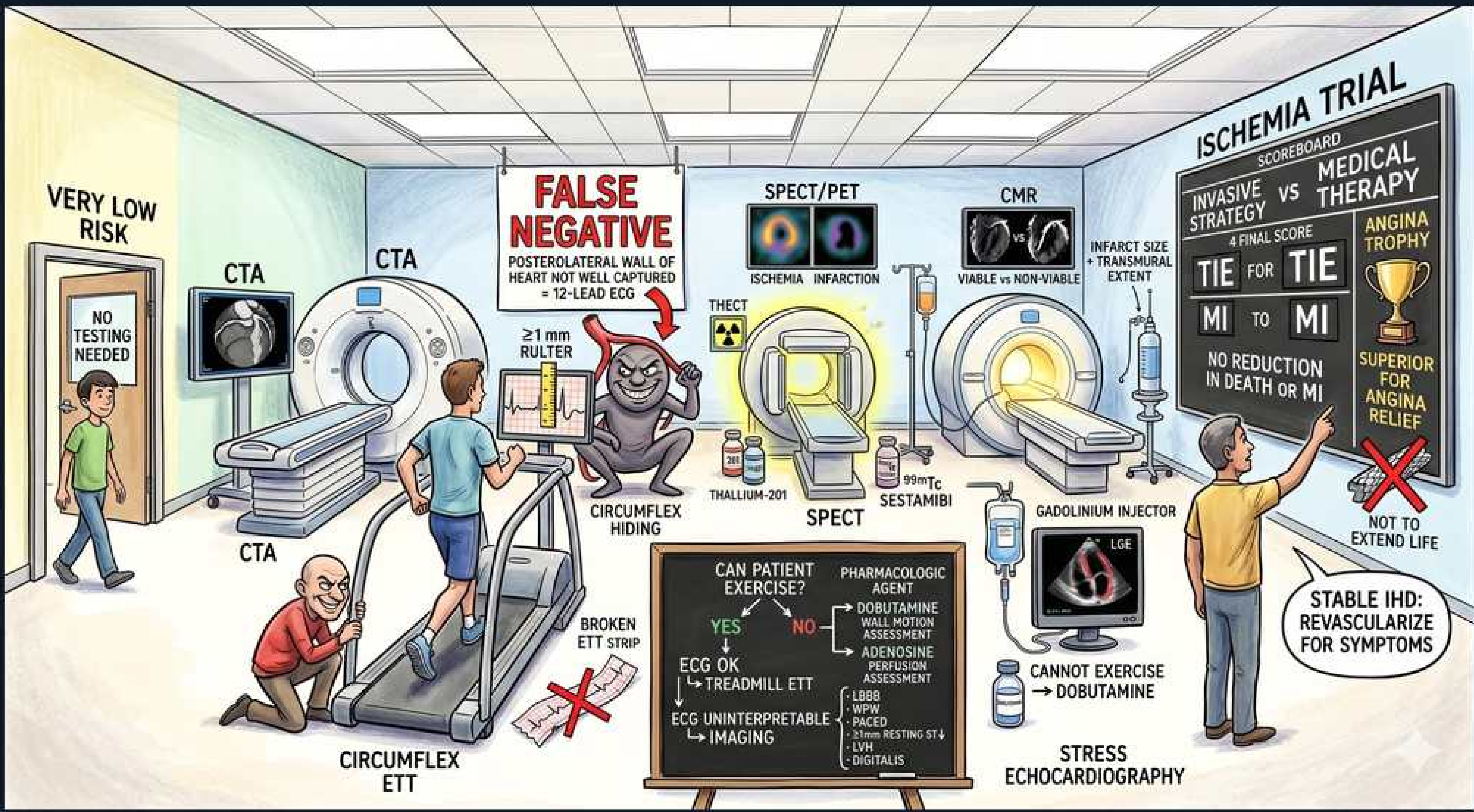
IHD — Stress Testing & CTA Decision Tree



1. Very low risk → no testing

WHAT YOU SEE

Green figure far left 'VERY LOW RISK / NO TESTING NEEDED'

WHAT IT ENCODES

In patients with very low pretest probability of CAD, no stress testing is needed — by Bayes' theorem a positive result is far more likely to be a false positive than true disease, so testing only generates harm (downstream angiography, anxiety, cost). Pretest probability is estimated from age, sex, and chest-pain character (typical vs atypical vs non-anginal). Reserve testing for intermediate pretest probability where it most changes management.

BOARD TRAP

WRONG: every patient with chest pain should get a stress test. Discriminator: at very low pretest probability, testing yields mostly false positives — it is the intermediate-risk patient who benefits most; over-testing the low-risk patient is the error.

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2. CTA (anatomic, rule-out)

WHAT YOU SEE

Left-side CT scanner panels each labelled 'CTA' displaying coronary-tree CT images — the anatomic camera that pictures the actual arteries to rule out CAD (excellent negative predictive value).

WHAT IT ENCODES

Coronary CT angiography is an ANATOMIC test with an excellent negative predictive value (>95%), making it ideal to RULE OUT CAD in low-to-intermediate pretest-probability patients — a normal CTA essentially excludes obstructive disease. It directly visualizes plaque and stenosis. Limitations: heavy calcium (high calcium score) and tachycardia/arrhythmia degrade images, and it gives anatomic, not functional/ischemic, information (often paired with FFR-CT).

BOARD TRAP

WRONG: coronary CTA is the preferred test in a patient with very high calcium burden or to assess functional ischemic significance. Discriminator: dense calcification blooms and obscures the lumen on CTA, and CTA shows anatomy not ischemia — use functional testing there, not CTA.

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3. Can patient exercise?

WHAT YOU SEE

Central chalkboard 'CAN PATIENT EXERCISE?' YES/NO branch

WHAT IT ENCODES

The first branch point in functional stress testing is: can the patient exercise AND is the baseline ECG interpretable? If YES to both → exercise treadmill ECG (cheapest, adds functional capacity data). If the patient cannot exercise → pharmacologic stress. If the ECG is uninterpretable → add imaging. Exercise is preferred whenever feasible because exercise capacity itself is prognostic.

BOARD TRAP

WRONG: pick the imaging modality first, then decide on exercise. Discriminator: the decision tree starts with 'can the patient exercise / is the ECG readable?' — modality follows from that branch; jumping to imaging first inverts the algorithm.

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4. YES → exercise ECG (treadmill)

WHAT YOU SEE

'YES → ECG OK → TREADMILL ETT'

WHAT IT ENCODES

If the patient can exercise AND the baseline ECG is normal/interpretable, exercise treadmill ECG (Bruce protocol) is the first-line, least-expensive functional test. It also yields prognostic data (exercise capacity in METs, BP response, Duke treadmill score) that imaging alone does not. Goal is ≥85% of age-predicted max HR for a diagnostic study.

BOARD TRAP

WRONG: order stress imaging up front in a patient who can exercise with a normal ECG. Discriminator: plain exercise ECG is first-line when feasible — adding imaging without an indication (uninterpretable ECG or inability to exercise) is unnecessary cost and over-testing.

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5. Exercise ETT (Bruce)

WHAT YOU SEE

Man running on treadmill, 'CIRCUMFLEX ETT', '≥1 mm RULER'

WHAT IT ENCODES

On exercise ECG, a positive result = ≥1 mm of horizontal or downsloping ST depression 80 ms after the J point during exertion. Upsloping ST depression is non-diagnostic. The Bruce protocol increases speed/grade every 3 minutes; the test is read for ST changes, symptoms, BP/HR response, and exercise duration (METs).

BOARD TRAP

WRONG: any ST depression (including upsloping) during exercise makes the test positive. Discriminator: only horizontal/downsloping ≥1 mm depression is positive — upsloping is non-diagnostic; the morphology, not mere presence of depression, defines positivity.

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6. Uninterpretable ECG → imaging

WHAT YOU SEE

'ECG UNINTERPRETABLE → IMAGING' (LBBB, paced, baseline ST)

WHAT IT ENCODES

When the baseline ECG is uninterpretable for ischemia — LBBB, ventricular paced rhythm, pre-excitation (WPW), LVH with strain, ≥1 mm resting ST depression, or digoxin effect — exercise ECG cannot be read and you must ADD IMAGING (stress echo or nuclear perfusion). Special case: with LBBB, exercise can cause false septal perfusion defects, so VASODILATOR (adenosine/regadenoson) nuclear stress is preferred over exercise or dobutamine.

BOARD TRAP

WRONG: rely on the exercise ECG alone when the baseline ECG shows LBBB or a paced rhythm. Discriminator: those baselines obscure ischemic ST changes — you must add imaging; and with LBBB specifically, choose vasodilator stress to avoid false septal defects.

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7. Cannot exercise → pharmacologic

WHAT YOU SEE

'CANNOT EXERCISE → DOBUTAMINE'

WHAT IT ENCODES

If a patient cannot exercise adequately (deconditioning, arthritis, PAD, amputation, COPD, neurologic deficit), use a PHARMACOLOGIC stress agent paired with imaging: a vasodilator (adenosine/regadenoson/dipyridamole) with perfusion imaging, or dobutamine with stress echo. Pharmacologic stress is mandatory here because a submaximal exercise test is non-diagnostic.

BOARD TRAP

WRONG: have the patient walk a submaximal treadmill test if they can't fully exercise. Discriminator: a submaximal exercise study (HR <85% predicted) is non-diagnostic — switch to pharmacologic stress with imaging rather than accept an inadequate exercise test.

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8. Adenosine/regadenoson (perfusion)

WHAT YOU SEE

'ADENOSINE → PERFUSION ASSESSMENT', vasodilator agent box

WHAT IT ENCODES

Vasodilators (adenosine, regadenoson, dipyridamole) dilate normal coronaries but not stenosed ones, creating heterogeneous flow ('coronary steal') that perfusion imaging (SPECT/PET) detects as a relative defect. Regadenoson is a selective A2A agonist given as a simple bolus and is better tolerated. Reverse adenosine/dipyridamole effects with aminophylline; avoid caffeine for 12–24h before the test as it blocks the response.

BOARD TRAP

WRONG: give vasodilator stress to a patient with active bronchospasm or high-grade AV block. Discriminator: adenosine/dipyridamole can trigger bronchospasm and AV block — contraindicated in severe asthma/COPD and 2nd/3rd-degree AV block without pacemaker; use dobutamine instead there.

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9. Dobutamine (wall motion)

WHAT YOU SEE

Dobutamine bottle, 'CANNOT EXERCISE → DOBUTAMINE' feeding stress echo

WHAT IT ENCODES

Dobutamine is a beta-1 agonist that increases heart rate and contractility (raising myocardial O2 demand) to provoke a WALL-MOTION abnormality on stress echocardiography when a patient cannot exercise. A new or worsening regional wall-motion abnormality = inducible ischemia. Atropine can be added to reach target HR; reverse excessive effects with a short-acting beta-blocker (esmolol).

BOARD TRAP

WRONG: dobutamine is the agent of choice for nuclear PERFUSION imaging. Discriminator: dobutamine is paired with stress ECHO (wall motion); vasodilators (adenosine/regadenoson) are the agents for nuclear PERFUSION imaging — matching agent to modality is the tested point.

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10. SPECT/PET perfusion

WHAT YOU SEE

'SPECT/PET' gamma-camera images, ischemia vs infarction

WHAT IT ENCODES

Nuclear perfusion imaging (SPECT with thallium-201 or Tc-99m sestamibi; PET with rubidium-82) distinguishes reversible ischemia from fixed infarct: a REVERSIBLE defect (present on stress, fills in at rest) = inducible ischemia; a FIXED defect (on both) = scar. PET offers higher resolution and quantitative myocardial blood flow but is less widely available.

BOARD TRAP

WRONG: a fixed perfusion defect indicates active ischemia that revascularization will improve. Discriminator: fixed = scar/infarct (non-viable); only reversible defects represent ischemia that benefits from revascularization.

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11. Stress echo / CMR viability

WHAT YOU SEE

'CMR' scanner, 'VIABLE = NON-VIABLE', infarct size / transmural extent

WHAT IT ENCODES

Stress echo detects inducible ischemia as new wall-motion abnormalities (no radiation, bedside, assesses valves/EF too). Cardiac MRI with late gadolinium enhancement assesses viability by scar transmurality — <50% wall-thickness scar is viable and recovers after revascularization, while transmural (>50%) late enhancement is non-viable and will not improve.

BOARD TRAP

WRONG: a segment with full-thickness (transmural) late gadolinium enhancement will regain function after revascularization. Discriminator: transmural scar = non-viable (no recovery); only subendocardial, limited-thickness scar is viable — revascularizing non-viable myocardium gives no functional benefit.

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12. False negative: circumflex

WHAT YOU SEE

'FALSE NEGATIVE' banner, gorilla 'CIRCUMFLEX HIDING', 12-lead note

WHAT IT ENCODES

The standard 12-lead ECG poorly represents the posterolateral wall supplied by the left circumflex artery (a relatively 'electrically silent' territory), so isolated circumflex disease is a classic source of FALSE-NEGATIVE exercise ECGs. Posterior leads (V7–V9) or imaging-based stress testing improve detection of this territory.

BOARD TRAP

WRONG: a normal exercise ECG reliably excludes circumflex (posterolateral) disease. Discriminator: the 12-lead underrepresents the posterolateral wall, so circumflex disease commonly produces false-negative exercise ECGs — a normal study does not exclude it.

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13. Stable IHD: revasc for symptoms

WHAT YOU SEE

Right banner 'STABLE IHD: REVASCULARIZE FOR SYMPTOMS, NOT TO EXTEND LIFE'; ISCHEMIA trial 'TIE for TIE'

WHAT IT ENCODES

The ISCHEMIA trial showed that in STABLE CAD with moderate-to-severe ischemia, an initial invasive strategy (routine revascularization) did NOT reduce death or MI versus optimal medical therapy — it improved angina/quality of life. So in stable IHD, revascularize for SYMPTOM relief, while optimal medical therapy (antiplatelet, high-intensity statin, BP control, antianginals, risk-factor modification) is the foundation. Exceptions where revascularization does improve survival: left main disease, and severe multivessel disease especially with reduced EF.

BOARD TRAP

WRONG: PCI prolongs life in stable IHD, so all patients with positive stress tests need revascularization. Discriminator: per ISCHEMIA, in stable CAD revascularization relieves angina but does not reduce death/MI vs medical therapy — survival benefit is reserved for left main/severe multivessel disease, not routine stable single-vessel disease.